Problem Set #1: Classical Mechanics (PH31207)

August 31, 2026

Introduction

1. Consider a reasonably isolated system of three masses m_1, m_2 and m_3. The radius vector of the ith particle from the origin is $\vec{r}_i$, where $i = 1, 2, 3$. Assume that the interaction between the masses obeys the Newton's law of gravitation. No external force is assumed to be acting on these masses.

   (a) Evaluate the summation

   $$\sum_{i=1}^{3} \sum_{\substack{k=1 \\ k \neq i}}^{3} \vec{r}_i \times \vec{F}_{ik}$$

   explicitly, where $\vec{F}_{ik}$ is the force on the ith mass due to the kth mass.

   (b) Is the total angular momentum conserved in this case?

2. Consider

   $$\vec{F} = \left(2xy + z^3\right) \boldsymbol{i} + \left(x^2 + 2y\right) \boldsymbol{j} + \left(3xz^2 - 2\right) \boldsymbol{k},$$

   where $\boldsymbol{i}, \boldsymbol{j}$ and $\boldsymbol{k}$ are unit vectors along x, y, and z directions respectively.

   (a) Find a scalar function $\phi(x, y, z)$ such that $\vec{F} = \vec{\nabla}\phi$. Show explicitly how you obtained ϕ.

   (b) Calculate $\vec{\nabla} \times \vec{F}$ separately.

   (c) Is the force conservative?

   (d) Consider a particle moving under the above force field in 3D. The equation for the trajectory of this particle is given by

   $$\vec{r}(t) = 3\boldsymbol{i} \cos(3t + \frac{\pi}{2}) + 3\boldsymbol{j} \sin(4t + \pi) - \boldsymbol{k}.$$

   How much work is done by the force on this particle in the time interval $[0, 2\pi]$? [Extra: Plot this curve; can you identify it?]